Robust State-Number Selection in Hidden Markov Models

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Abstract: Hidden Markov models (HMMs) are widely used to infer latent behavioral states from animal movement and biologging time series, but selecting the number of states is difficult when fitted models only approximate complex processes. Under misspecification, the Bayesian information criterion (BIC) can favor increasingly complex models as data grow, while the integrated completed likelihood (ICL) behaves inconsistently across settings. We adapt the *accumulated cutoff discrepancy criterion* (ACDC) for HMM state-number selection and develop an improved approach to choosing its cutoff. In a simulated spike-train experiment, ACDC recovers the true five-state structure despite overdispersion and burst contamination, whereas BIC and ICL overfit. In analyses of elephant GPS movement under three angular-emission models and canine biologging data with overlapping behavior annotations, ACDC identifies stable, interpretable state decompositions. BIC favors larger models that absorb residual emission or transition mismatch, and ICL gives inconsistent recommendations. Our results demonstrate that ACDC is a more reliable and robust method of model selection for HMMs than current state-of-the-art approaches.

Keywords and phrases: Accumulated cutoff discrepancy criterion; animal movement; biologging; hidden Markov models; model misspecification; time-series analysis.

1. Introduction

Hidden Markov models (HMMs) have become a standard tool in animal movement and biologging research for inferring latent behavioral states from time series of telemetry-derived movement features and high-frequency sensor measurements (Langrock et al., 2012; McClintock and Michelot, 2020; Patterson et al., 2017; Zucchini, MacDonald, and Langrock, 2016). By modeling observed data as emissions from an unobserved discrete-state process, HMMs provide a flexible framework for summarizing complex animal behaviors into a small number of interpretable movement or activity modes (Hooten et al., 2017).

A central modeling decision in applied HMM analyses is the choice of the number of latent states (Pohle et al., 2017; Zucchini et al., 2016). The selected number of states directly determines model complexity and strongly influences downstream inference, including state decoding, behavioral interpretation, and subsequent ecological conclusions. In practice, state-number selection in movement HMMs is most commonly performed using likelihood-based information criteria such as the *Akaike information criterion* (AIC) or the *Bayesian information criterion* (BIC), often supplemented by informal heuristics such as scree plots, elbow rules, and subjective assessments of biological interpretability (Pohle et al., 2017). While these pragmatic approaches are widely used, they lack a principled statistical foundation for settings in which the fitted HMM is only an approximation to the underlying data-generating process.

Model misspecification is pervasive in animal movement and biologging applications. Emission distributions are typically simplified representations of complex, heterogeneous behavioral signals, and the latent dynamics are often forced into a first-order, time-

homogeneous Markov form with geometric dwell times (Langrock and Zucchini, 2011) even when real behavior exhibits longer memory, regime persistence, or task-dependent transition structure. In addition, temporal dependence structures may be only partially captured, and unmodeled covariates or individual-level variability are common (Glennie, Adam, and Leos-Barajas, 2023). Under such misspecification, likelihood-based information criteria can exhibit unstable behavior, often favoring increasingly large numbers of latent states as sample size grows, with additional states absorbing modeling error rather than representing meaningful behavioral structure (Cai, Campbell, and Broderick, 2021; Miller and Dunson, 2019; Pohle et al., 2017). This phenomenon complicates the use of HMMs as exploratory tools for latent behavior discovery, particularly in large modern telemetry and sensor datasets.

Several alternatives to classical information criteria have been proposed for selecting the number of states in HMMs. These include cross-validation-based approaches (Celeux and Durand, 2008), marginal likelihood or evidence-based methods (Chen, Fuh, and Kao, 2025), and Bayesian nonparametric formulations that treat the number of states as unknown (Beal, Ghahramani, and Rasmussen, 2002; Fox et al., 2011). These methods can offer stronger theoretical guarantees under idealized assumptions but also require more computation or more complex inference procedures. In applied movement ecology, the *integrated completed likelihood* (ICL) has also been considered as a criterion for state-number selection, providing a likelihood-based alternative to BIC that incorporates uncertainty in state assignments (Pohle et al., 2017). Despite these developments, there remains a practical need for state-number selection methods that are both easy to integrate into existing HMM pipelines and robust to the forms of model misspecification encountered in movement and biologging data.

In this article, we address this gap by extending the *accumulated cutoff discrepancy*

criterion (ACDC) of Li et al. (2026), a recently developed model selection criterion designed to remain interpretable under model misspecification, to the problem of state-number selection in HMMs. ACDC is a drop-in alternative to classical information criteria, requiring no modification to standard estimation procedures and little additional computational overhead. This chapter also uses an envelope visualization for the ACDC cutoff that aligns with the "elbow plots" familiar to applied users. We evaluate the behavior of ACDC on empirical animal-movement datasets spanning both location-based movement features and accelerometer-derived behavioral signals, as well as a simulated spike-train benchmark with known latent structure, and compare its performance against BIC and ICL. Our results demonstrate that ACDC yields more stable and interpretable state-number selections in applied movement settings and recovers the correct state number under simulated misspecification.

The remainder of this article is organized as follows. In Section 2, we formalize the HMM state-selection problem, then review the likelihood-based BIC and ICL baseline methods. In Section 3, we introduce ACDC, extend it to HMMs, and develop an envelope visualization for selecting the ACDC cutoff. Section 4 presents our experimental results. In Section 5 we conclude by discussing the implications of our findings and directions for future work.

2. State-number selection criteria

2.1. Hidden Markov model formulation and selection task

A *K-state hidden Markov model* (HMM) specifies the joint distribution of an observed time series $y_{1:T} = (y_1, \ldots, y_T)^{\mathrm{T}}$, with $y_t \in \mathbb{Y} \subseteq \mathbb{R}^D$, together with an unobserved *state sequence*

$z_{1:T} = (z_1, \ldots, z_T)^{\mathrm{T}}$ taking values in $\{1, \ldots, K\}$. The states form a first-order Markov chain: z_1 is drawn from an initial distribution $\delta^{(K)} = (\delta_1, \ldots, \delta_K)$, and $\mathbb{P}(z_{t+1} = j \mid z_t = i) = \gamma_{ij}$ for a $K \times K$ transition matrix $\Gamma^{(K)} = (\gamma_{ij})$. Conditional on the states, the observations are independent, with y_t given $z_t = k$ drawn from a state-dependent *emission distribution* $F_{K,k}$ having density $f_{K,k}$ and parameter ϕ_k; for instance, $F_{K,k} = \mathcal{N}(\mu_k, \Sigma_k)$ in a Gaussian-emission HMM. The model is therefore parameterized by $\theta_K = (\delta^{(K)}, \Gamma^{(K)}, \phi_1, \ldots, \phi_K)$, and the complete-data density factorizes as

$$p_{\theta_K}(y_{1:T}, z_{1:T}) = \delta^{(K)}_{z_1} \prod_{t=2}^{T} \gamma_{z_{t-1} z_t} \prod_{t=1}^{T} f_{K,z_t}(y_t), \tag{1}$$

with marginal likelihood $p_{\theta_K}(y_{1:T}) = \sum_{z_{1:T}} p_{\theta_K}(y_{1:T}, z_{1:T})$ (Rabiner, 1989; Zucchini et al., 2016). We assume that, for each candidate K value, the transition matrix $\Gamma^{(K)}$ is irreducible and aperiodic, so the chain is ergodic and has a unique stationary distribution $\pi^{(K)}$.

Let $\mathcal{K}$ denote the finite set of candidate state numbers considered by the analyst, for example $\mathcal{K} = \{1, \ldots, 10\}$. For each $K \in \mathcal{K}$, we fit an HMM by maximum likelihood and denote the fitted parameter vector by $\hat{\theta}_K$. Our goal is to choose K so that the fitted model captures interpretable latent behavioral structure without introducing spurious states that only explain the misspecification. To formalize this, let

$$\ell_K = \log p_\theta(y_{1:T})|_{\theta=\hat{\theta}_K} \tag{2}$$

be the maximized marginal log-likelihood and let d_K be the number of free parameters in the K-state model. The criteria considered below assign each candidate K a scalar score based on fit, complexity, and (for ICL) state-assignment uncertainty; the selected state number is the minimizer of that score over $\mathcal{K}$.

2.2. Bayesian Information Criterion

We use the *Bayesian information criterion* (BIC) (Schwarz, 1978) as a likelihood-based baseline:

$$\mathrm{BIC}(K) = -2\ell_K + d_K \log T, \tag{3}$$

where T is the number of observations. We omit the *Akaike information criterion* (AIC) (Akaike, 1974) because its smaller complexity penalty, $2d_K$, cannot select fewer states than BIC for the sample sizes considered here ($T > e^2$).

For movement and biologging HMMs, BIC remains attractive because it is easy to compute from standard fitting outputs and scales well to large datasets. Under misspecification, however, likelihood gains from extra states can outweigh BIC's penalty, favoring states that absorb residual heterogeneity rather than distinct behavioral modes.

2.3. Integrated Completed Likelihood

The *integrated completed likelihood* (ICL; Biernacki, Celeux, and Govaert, 2000) augments likelihood-based selection by accounting for uncertainty in latent-state assignments. For HMMs, a common practical form is

$$\mathrm{ICL}(K) = \mathrm{BIC}(K) + 2\,\mathcal{H}_K, \tag{4}$$

where the entropy term $\mathcal{H}_K$ is computed from the posterior state probabilities $\hat{\gamma}_{tk}^{(K)} = \mathbb{P}_{\hat{\theta}_K}(z_t = k \mid y_{1:T})$ as

$$\mathcal{H}_K = -\sum_{t=1}^{T}\sum_{k=1}^{K} \hat{\gamma}_{tk}^{(K)} \log \hat{\gamma}_{tk}^{(K)}, \tag{5}$$

with the convention $0 \log 0 = 0$. For multiple independent tracks or trials, the outer sum is taken over all observation times in all sequences. When posterior state membership is

diffuse, $\mathcal{H}_K$ is larger, increasing the criterion value and discouraging models that achieve fit by introducing poorly separated states; when the decoded states are nearly certain, $\mathcal{H}_K$ is close to zero and ICL is close to BIC.

Therefore, relative to BIC, ICL tends to prefer solutions with clearer state separation and more stable decoding, which can be beneficial in behavioral interpretation tasks. At the same time, ICL may under-select K when biologically meaningful states overlap substantially in observation space.

3. Accumulated Cutoff Discrepancy Criterion (ACDC) for State-number Selection

The *accumulated cutoff discrepancy criterion* (ACDC) was introduced by Li et al. (2026) as a robust model-selection procedure for latent-variable models under misspecification. The motivating observation is that, in explanatory settings, the goal is not merely to optimize predictive fit but to recover a parsimonious collection of latent components that still correspond to meaningful underlying processes. When the model family is misspecified, a predictive discrepancy such as KL divergence or expected log loss typically decreases monotonically as additional states are added, so criteria based on the likelihood can keep rewarding extra states even after the scientifically relevant latent structure has already been captured. ACDC addresses this issue by shifting the goal from *overall predictive fit* to *component-level adequacy*: whether each fitted latent component is an accurate approximation to a distinct process up to a user-specified tolerance.

In the general formulation, one imagines an unknown component-level discrepancy between the data-generating mechanism and each component of a fitted K-component model. If the

tolerated discrepancy level ρ were known, the ideal target would be the smallest K for which all fitted components are adequate at that tolerance. Because the population quantities are unavailable, ACDC replaces them by plug-in estimates computed from the empirical distribution and fitted model parameters. Let $\widehat{D}_{K,k}$, which will be defined precisely below, denote the estimated discrepancy for component $k \in \{1, \ldots, K\}$ in a fitted K-state HMM. For a fixed cutoff $\rho \geq 0$, ACDC defines the loss

$$R_\rho(K) = \sum_{k=1}^{K} \max\bigl(0, \widehat{D}_{K,k} - \rho\bigr). \tag{6}$$

The truncation at ρ means that discrepancies below the tolerated misspecification level are treated as acceptable, while only excess discrepancy contributes to the loss. The loss may be minimized by more than one candidate state number, so we first define the minimizing set

$$\mathcal{M}_\rho = \arg\min_{K \in \mathcal{K}} R_\rho(K). \tag{7}$$

The selected state number is then the smallest element of this set,

$$\widehat{K}_\rho = \min \mathcal{M}_\rho. \tag{8}$$

3.1. Application to HMMs

To apply ACDC to an HMM, we must identify its latent components and a discrepancy that measures their adequacy. Taking the chain to be stationary, a single observation follows the mixture law

$$y_t \sim \sum_{k=1}^{K} \pi_k^{(K)} \, F_{K,k}, \tag{9}$$

so that, marginally, a K-state HMM is a K-component mixture whose weights are the stationary state probabilities $\pi^{(K)}$ and whose components are the emission distributions

$F_{K,k}$. Each fitted state therefore plays the role of a mixture component, and its adequacy is judged by comparing the fitted emission $F_{K,k}$ to the distribution of the observations the model assigns to that state.

Concretely, let $\hat{\gamma}_{tk}^{(K)} = \mathbb{P}_{\hat{\theta}_K}(z_t = k \mid y_{1:T})$ be the smoothed posterior probability that observation t was generated by state k, which are the same responsibilities that appear in ICL. Weighting each observation y_t in proportion to $\hat{\gamma}_{tk}^{(K)}$ yields the inferred state-k distribution

$$\widehat{F}_{K,k,T} = \frac{\sum_{t=1}^{T} \hat{\gamma}_{tk}^{(K)} \delta_{y_t}}{\sum_{t=1}^{T} \hat{\gamma}_{tk}^{(K)}}. \tag{10}$$

The finite-sample plug-in discrepancy estimates the KL divergence between the limiting state-assigned distribution, defined formally in Web Appendix A, and the fitted emission,

$$D_{K,k}^{\mathrm{KL}} = D_{\mathrm{KL}}\big(F_{K,k}^{\star} \,\|\, F_{K,k}\big)\,, \tag{11}$$

which is small precisely when $F_{K,k}$ accurately models the observations assigned to state k. Because $\widehat{F}_{K,k,T}$ is a normalized posterior-weighted empirical distribution, we estimate $D_{K,k}^{\mathrm{KL}}$ with the k-nearest-neighbor estimator of Li et al. (2026) (based on Kozachenko and Leonenko, 1987; Wang, Kulkarni, and Verdú, 2009), which combines a nearest-neighbor entropy estimate for the state-k observations with the emission density $f_{K,k}$; the resulting nonnegative value is the plug-in $\widehat{D}_{K,k}$ entering the ACDC loss.

3.2. Theoretical Foundations

We now establish that, even under misspecification, ACDC can recover the number of HMM states, K_o, as the sample size grows. The difficulty is that because every candidate HMM may be misspecified, the data-generating distribution P_o can lie outside each fitted family.

Classical model-selection consistency is unattainable in this regime, since it requires the true model to belong to the fitted family. The relevant guarantee is instead *robust model selection consistency*, which Li et al. (2026) establish for mixture models with independent observations; we extend it to HMMs, specializing throughout to the KL divergence for simplicity.

Robust model selection consistency replaces correct specification by a tolerance on the per-component fit. Let ρ_o denote the boundary component-level discrepancy at the true state number—the largest population value of the KL divergence $D^{\mathrm{KL}}_{K_o,k}$ defined in equation 11, taken over the components k and over the limiting parameters of the K_o-state fit. A selection rule is *robustly consistent* if it recovers K_o whenever the fitted models for smaller K are separated from P_o: for every $K < K_o$,

$$\inf_{\theta \in \Theta_\star^{(K)}} d(P_o, P_\theta) > \kappa(\rho, K), \tag{12}$$

where P_θ is the marginal observation law for K states with parameter θ, $\Theta_\star^{(K)}$ is the set of limiting parameters of the K-state fit, d is a metric on distributions, and $\rho > \rho_o$ is the cutoff used by the selection rule. The *gap function* κ sets how much worse a coarser model must be before it is rejected. Condition (12) is the price of misspecification: a model with too few states is discarded only when it is meaningfully farther from P_o than the K_o-state model, with the required margin governed by the tolerance ρ.

Casting HMM state selection in this framework uses the marginal mixture representation. As shown in equation 9, a stationary K-state HMM is marginally a K-component mixture whose components are the emissions. The component-level discrepancy $D^{\mathrm{KL}}_{K,k}$ is the corresponding HMM analogue of the mixture-model discrepancy of Li et al. (2026):

it measures how far the inferred distribution of the observations attributed to state k is from the fitted emission $F_{K,k}$. The main structural difference from an independent mixture is the Markov dependence among the states. The discrepancy still compares state-specific emission distributions, but the inferred state-k distribution in equation 10 is posterior-weighted rather than an ordinary empirical distribution. Thus the HMM argument requires three convergence inputs. First, the unweighted empirical distribution $T^{-1}\sum_{t=1}^{T}\delta_{y_t}$ converges by the ergodic theorem for a stationary, ergodic observation process. Second, the fitted HMM parameters must stabilize as T grows. Existing HMM consistency results give this under standard regularity conditions: even when the fitted HMM family is misspecified, $\hat{\theta}_K$ converges to a limiting parameter $\theta_K^\star$ that minimizes the KL divergence rate between the true observation process and the K-state HMM family (Douc and Moulines, 2012; Douc et al., 2011; Leroux, 1992). Third, because ACDC uses posterior state weights to form the state-specific empirical distributions, those weights must have a stable limiting behavior. A sufficient set of regularity conditions, together with the posterior-weighted convergence argument, is given in Web Appendix A. We therefore obtain the following guarantee.

Theorem 1 (Robust consistency for HMM state selection)**.** *Suppose ACDC is applied with the component-level KL divergence of equation 11. Assume that, for each candidate $K \le K_o$, the maximum-likelihood estimator $\hat{\theta}_K$ is consistent and the regularity conditions of Web Appendix A hold. Then ACDC is robustly consistent with gap function $\kappa(\rho, K) = \sqrt{\rho/2}$: whenever $\rho > \rho_o$ and P_o and K_o satisfy the separation condition (12) for this κ, the selected state number satisfies $\mathbb{P}(\widehat{K}_\rho = K_o) \to 1$ as $T \to \infty$.*

In words, Theorem 1 guarantees that, despite the misspecification of the fitted HMMs,

ACDC can recover the smallest state number beyond which every latent component is already adequate at the tolerated discrepancy level. The proof is given in Web Appendix A.

3.3. An Improved Cutoff Selection Procedure

The choice of the cutoff value ρ is application dependent. When related labeled datasets or realistic simulations are available, Li et al. (2026) suggest calibrating ρ empirically from downstream performance. When such calibration data are unavailable, they propose examining the loss curves $R_\rho(K)$ across ρ and selecting the model that is optimal over the first substantial interval of ρ values, with ρ kept as small as possible. This rule favors the first stable model, before larger values of ρ begin to make overly coarse models acceptable. However, the loss curves $R_\rho(K)$ that Li et al. (2026) plot against ρ can be hard to read, since they show a separate curve for each value of K; the clutter only worsens as the maximum K grows.

To address this challenge, we propose to evaluate the optimal value of ρ using the same heuristic, but using a simpler visualization we call the ACDC *envelope plot*. For each candidate K, we first compute the worst state-level discrepancy

$$\widehat{D}^*_K = \max_{1\le k\le K} \widehat{D}_{K,k}, \tag{13}$$

using the best fit for that criterion. The plotted envelope is the monotone lower envelope of the curve $\widehat{D}^*_K$ over K, retaining only the candidate state numbers that improve on all smaller models. In Figure 1, for example, the envelope falls sharply as K approaches the ground-truth five-state model, while later envelope steps are much smaller. The envelope therefore turns the original stability-over-ρ heuristic into an elbow diagnostic, looking for candidates that give a

large improvement over previous envelope points, after which subsequent improvements are at most marginal. This elbow-style diagnostic is more familiar and easier for practitioners to use.

As we will see empirically, the elbow point for ACDC is consistently more clear than the elbow point for BIC or ICL. Intuitively, this is because the BIC and ICL average over state-level fits, so including additional states tends to have a smoothly diminishing effect in the misspecified setting. ACDC elbow plots, on the other hand, is based on worst-case, component-level discrepancy, which can change significantly even with the addition of only a single new state. Moreover, the ACDC discrepancy is the KL-divergence, so the y-axis of the ACDC elbow plots are interpretable, whereas they are more difficult to interpret in the BIC/ICL case.

4. Applications

We evaluate state-number selection methods on three benchmarks representing common data modalities and modeling regimes in behavioral time-series analysis: a synthetic spike-train experiment that contrasts well-specified and misspecified count emissions, an elephant movement dataset based on GPS-derived trajectories, and a large-scale canine biologging dataset with high-frequency inertial sensor measurements. For each candidate state number, we fit multiple randomly initialized HMMs rather than relying on a single EM solution. Unless otherwise noted, all fitted models are retained and each selection criterion is evaluated using its own best-scoring fit at that candidate state number, so that BIC, ICL, and ACDC are each compared on the solution most favorable to their respective objective.

4.1. Synthetic spike-train experiment

We begin with a simulated spike-train benchmark as a proof of concept, because the latent state structure is known by construction. Unlike the empirical movement datasets, this setting lets us directly check whether the selection criteria recover the ground-truth number of states. Poisson-emission HMMs of this kind are a standard model for multineuronal spike-train activity (Escola et al., 2011).

4.1.1. Experiment design

We fit Poisson-emission HMMs over $K = 1, \ldots, 10$ under two data-generating regimes. In the well-specified case, the data are generated from a five-state Poisson HMM observed across five simulated neurons, using $T = 20{,}000$ bins of width 100 ms and sticky transition dynamics, where the self-transition probabilities P_{ii} are close to one. The ground-truth states consist of a low-rate silent state, a chaotic mixed-rate state, a higher-rate up state, and two burst states with distinct neuron-specific firing patterns; Table 1 lists the corresponding rates and realized occupancies. In the misspecified case, we keep the same latent state sequence and state-specific mean firing rates but alter the observations in two ways. First, non-contaminated bins are generated from negative-binomial emissions with the same means as the Poisson model and variance equal to twice the mean:

$$X_{td} \mid z_t = s \sim \mathrm{NegBin}(r = \mu_{sd}, p = 0.5),$$

where $\mu_{sd} = 0.1\lambda_{sd}$ is the expected count for neuron d in state s during a 100 ms bin. Thus, the fitted Poisson HMM has the correct conditional mean but cannot represent the within-state overdispersion. Second, we add rare bursty contamination: with probability 0.01 when

no burst is active, a 10-bin burst begins and observations are drawn from one of 10 fixed Poisson contamination modes whose neuron-specific rates are generated independently of the ground-truth states from a log-uniform distribution on $[0.5, 50]$ Hz. The latent state sequence is unchanged; the contamination affects only the observed counts. In the realized sequence, 1,770 of 20,000 bins, or 8.85%, are contaminated. This design tests whether the criteria recover the ground-truth firing regimes or instead allocate additional states to overdispersion and off-structure burst modes.

4.1.2. Empirical findings and criterion comparison

The selection diagnostics in Figure 1 show correct recovery in the well-specified case, with BIC, ICL, and ACDC all identifying the ground-truth five-state model. The accompanying composition diagnostics confirm that the selected five-state fit recovers the intended state structure, with each firing regime mapped almost one-to-one to a fitted component.

The misspecified case demonstrates the robustness benefits of ACDC, with an elbow plot that shows a large decrease from $K = 4$ to $K = 5$. The BIC and ICL elbow plots lack such a clear cut-off at $K = 5$. Moreover, selecting the best fitted model results in BIC and ICL overfitting. Web Figures 1 and 2 show representative high-K fits. The five-state fit still recovers the ground-truth firing regimes, whereas the additional high-K states account for misspecification rather than capture real latent structure.

4.2. Elephant movement dataset

4.2.1. Dataset description

As a second benchmark, we use an individual-level GPS movement dataset from the Movebank movement-syndrome vertebrate tracking repository (Abrahms et al., 2017). We analyze elephant 8, the most-observed individual in the repository, which has 15,957 GPS fixes (Web Figure 3). This choice provides enough observations for per-animal HMM fitting while avoiding heterogeneity from pooling multiple individuals. We convert the raw latitude-longitude fixes into two standard movement features (Morales et al., 2004): step length, measured by great-circle distance between consecutive fixes, and turn angle, measured by the change in bearing across triples of fixes. After removing endpoint observations and missing locations, the resulting sequence contains 15,955 valid step-angle observations.

4.2.2. Emission specifications

Because the elephant GPS data have no ground-truth behavioral labels, we cannot validate a selected state number by direct recovery of known states. Instead, we use this dataset as a robustness check across emission families with different levels of suitability and expressivity. All models use the same two movement features, log-step length and turn angle, but differ in how they represent the angular component.

We compare three emission specifications that frame the same features differently. The baseline pairs a Gaussian log-step model with a wrapped Cauchy turn-angle model (Mardia and Jupp, 2000). The alternatives are (1) a full-covariance Gaussian on log-step and raw turn angle, which breaks circular topology by separating angles near $-\pi$ and π, and (2) a

Gaussian log-step model with a cardioid angle model, which is circular but less expressive. If a criterion recommends the same state number across these framings, that consistency suggests it is detecting an underlying movement pattern rather than an artifact of one emission specification.

4.2.3. Empirical findings

The three emission specifications give the same overall result (Figure 2): BIC selects the largest tested model, ICL selects a two-state model, and ACDC recommends a stable decomposition at $K = 3$. The BIC result is consistent with continued likelihood gains from adding states, including under the Gaussian raw-angle specification that is grossly misspecified for circular turn angles. There is no clear elbow in any of the BIC plots. ICL is more robust and returns a defensible coarse split between near-stationary and moving behavior (Web Figure 4). ACDC supports three states all three specifications, with little-to-no improvement for $K \geq 4$. The three-state decomposition is interpretable, corresponding to near-stationary movement, medium-speed movement, and fast directed travel (Figure 3). In the Gaussian raw-angle specification, ACDC also shows a secondary high-K improvement at $K = 8$ (Figure 2b). The comparison does not identify a single correct state number in the absence of ground-truth labels. Rather, it shows that both ICL and ACDC give stable recommendations across models with very different geometric compatibility and expressivity, albeit at different levels of conservativeness. Web Appendix C provides fits at $K = 4$ (Web Figure 5) and at the BIC-favored $K = 8$ (Web Figures 6–8) for all three specifications.

4.3. Dog behavior biologging dataset

4.3.1. Dataset description

The final and most complex benchmark is a canine behavior dataset that contains high-frequency inertial measurements from wearable sensors on domestic dogs together with task labels and fine-grained behavioral annotations (Vehkaoja et al., 2025). Hidden Markov models are widely used to infer behavior from such accelerometer data (Leos-Barajas et al., 2017). The annotations can overlap within the same time interval, so the latent states need not correspond one-to-one with a single observed behavior label. This combination of rich sensor measurements and partially overlapping behavioral structure makes the dataset a useful test of state-number selection in a realistic misspecified setting.

4.3.2. Feature construction and bout segmentation

We constructed a multi-sequence HMM benchmark from the dataset. We focus on a balanced set of task bouts from three dogs, with four to five bouts from each of seven task categories: walking, sitting, trotting, lying down, playing, standing, and treat-search. Each bout is treated as an independent sequence drawn from a shared HMM, so the fitted transition matrix describes within-bout dynamics rather than artificial transitions across unrelated task periods. The resulting training set contains 29 task bouts and 52,174 non-overlapping windows. We aggregate raw timesteps into short non-overlapping windows to reduce the number of HMM observations for computational tractability and to define within-window standard deviations as motion-variability features. In this analysis, we use windows of 10 raw timesteps.

For this experiment, the back- and neck-mounted inertial signals are first averaged channel-by-channel. Within each window, we then compute per-axis means and standard deviations for the averaged tri-axial accelerometer and gyroscope channels, yielding a 12-dimensional observation vector. Retaining the spatial axes separately is important because body orientation, posture, and motion amplitude are encoded in the axis-specific gravity projection and variance structure. The full decoded timeline contains 69,372 windows; we use it for behavior-composition and behavior-pair summaries after fitting, while the HMM likelihood itself is computed on the bout sequences. We fit full-covariance Gaussian-emission HMMs for $K = 1, \ldots, 14$ on these bout sequences, using 20 random restarts for each candidate state number.

4.3.3. State interpretation via behavior pairs

We summarize each fitted state by the empirical distribution of behavior pairs observed in the windows assigned to that state. Specifically, each raw timestep can contain up to three concurrent behavior annotations, so for every window we identify the two most prevalent behavior codes within that window and treat them as a behavior pair. Windows with only one observed behavior are encoded as `behavior+(pure)`. Aggregating these pair counts within each latent state provides a more granular description than assigning a single dominant behavior label. This is important because many windows contain concurrent behaviors, such as posture plus locomotion or panting, so a one-label summary can hide the mixed annotations that explain why a state is used.

4.3.4. Qualitative empirical findings

The canine bout-HMM experiment provides a clear example of ACDC identifying an intermediate model rather than either the coarsest interpretable partition or the largest likelihood-improving model. The selection diagnostics in Figure 4 show that BIC and ICL both select $K = 13$. Using an elbow-based approach for BIC and ICL is difficult due to the smooth, gradual decay of the scores. By contrast, the ACDC envelope shows a substantial improvement from $K = 4$ to $K = 5$, after which the improvements are negligible.

The earlier envelope points at $K = 1$, $K = 2$, $K = 3$, and $K = 4$ correspond to coarse separations of broad activity levels, whereas the improvement from $K = 4$ to $K = 5$ separates high-rate locomotion from the head-down and slow-ambulation regime. After $K = 5$, each envelope improvement is small, so the larger models are best read as refinements rather than new structural decompositions. Table 2 summarizes the five states in the $K = 5$ fit. The selected model distinguishes a combined head-down and slow-ambulation regime, low-motion posture states, ballistic whole-body motion, panting-associated rest, and high-rate locomotion. The state labels are not one-to-one behavior labels. Instead, they summarize recurring sensor regimes that can be interpreted through the behavior-pair annotations. This interpretation is supported by the behavior-pair visualization in Figure 5.

The larger-K fits show what is gained and lost after the ACDC-selected model. At $K = 6$, the head-down and slow-ambulation state begins to split, separating sniffing and drinking more cleanly from walking and eating windows. At $K = 7$, the still-posture regime splits further along a small gyroscopic-stability axis, but at the cost of some states (e.g., sit + pant) being a mixture of several behaviors. While $K = 6$ is defensible, its benefits are

smaller than those of the $K = 4$ to $K = 5$ transition and do not change the main sensor-regime decomposition. Moreover, $K = 7$ comes at the cost of capturing some latent states less accurately, so it is not clearly superior to $K = 5$ or $K = 6$. For $K \geq 8$, the envelope improvements remain small and several intermediate values fall off the envelope, indicating that the extra components mainly redistribute residual error.

5. Discussion

ACDC reframes state-number selection around component-level adequacy rather than overall predictive fit. It seeks the smallest number of states at which each state sufficiently describes a distinct behavioral process. Across the three benchmarks, ACDC yielded stable and interpretable state counts in misspecified HMMs, precisely the setting where likelihood-based criteria are most vulnerable to residual structure. This kind of robust model selection matters most in the scientific-discovery setting that motivates the method, where latent states need not correspond one-to-one with observed labels. The canine analysis illustrates this point. Its annotations are informative but overlap within windows, so the fitted states are interpreted through sensor and behavior-pair summaries rather than matched to single behaviors. When no ground-truth count exists, the operative target is the coarsest resolution that remains behaviorally interpretable, which is exactly what component-level adequacy is designed to identify. BIC selects ever-larger models as the likelihood keeps improving, whereas ICL varies across settings, selecting a conservative model for the elephant data but tracking BIC in the canine analysis. That ACDC remains stable across all three settings reflects the robust-consistency guarantee of Theorem 1.

Several limitations and precautions should be noted. First, ACDC is applied after

model fitting, so its behavior depends on obtaining sufficiently consistent HMM fits. The EM algorithm could thus be replaced by a possibly more stable approach, such as a Bayesian posterior approximation. Second, the choice of ρ remains heuristic without external calibration data, motivating more systematic calibration rules for applications where the envelope is less clear than in the examples studied here; see, for example, Li et al. (2026); Xue et al. (2026). Third, ACDC is a model selection tool rather than a substitute for scientific interpretation. After a state number is recommended, the fitted states still need to be examined through emission summaries, decoded usage, and domain-specific annotations.

There are a number of directions for future work. We focused on finite-state HMMs, but the same approach could extend to other dynamic latent-state models, such as latent factor models. Future research could also consider ACDC in a Bayesian decision-theoretic setting. Bayesian HMMs have applications beyond animal trajectory analysis; for example, many dynamic capture-recapture models can be specified as HMMs, and interpretability can motivate a Bayesian framework (e.g., Gimenez, 2026). In that setting, ACDC quantities such as $D_{K,k}$ could be treated as derived quantities for posterior inference. A Bayesian treatment would face the same uncertainty-propagation issues as other model-selection approaches (Hooten and Hobbs, 2015), but ACDC could provide a useful target for Bayesian model selection under misspecification.

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material into the paper.

Data Availability

The elephant GPS observations are available from the Movebank Data Repository at `https://doi.org/10.5441/001/1.hm5nk220` (Abrahms et al., 2017). The canine movement sensor observations are available from Mendeley Data, version 4, at `https://doi.org/10.17632/vxhx934tbn.4` (Vehkaoja et al., 2025). The synthetic spike-train observations were generated for this study according to the design in Section 4.1.

References

Abrahms, B., Seidel, D. P., Dougherty, E., Hazen, E. L., Bograd, S. J., Wilson, A. M., et al. (2017). Data from: Suite of simple metrics reveals common movement syndromes across vertebrate taxa. Movebank Data Repository. `https://doi.org/10.5441/001/1.hm5nk220`.

Akaike, H. (1974). A new look at the statistical model identification. *IEEE Transactions on Automatic Control* **19,** 716–723.

Beal, M. J., Ghahramani, Z., and Rasmussen, C. E. (2002). The infinite hidden Markov model. In *Advances in Neural Information Processing Systems*, volume 14, pages 577–584.

Biernacki, C., Celeux, G., and Govaert, G. (2000). Assessing a mixture model for clustering with the integrated completed likelihood. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **22,** 719–725.

Cai, D., Campbell, T., and Broderick, T. (2021). Finite mixture models do not reliably learn the number of components. In *Proceedings of the 38th International Conference*

on Machine Learning, volume 139 of *Proceedings of Machine Learning Research*, pages 1158–1169. PMLR.

Celeux, G. and Durand, J.-B. (2008). Selecting hidden Markov model state number with cross-validated likelihood. *Computational Statistics* **23,** 541–564.

Chen, Y.-C., Fuh, C.-D., and Kao, W.-L. (2025). Determining the number of states in hidden markov models via marginal likelihood. *Journal of Machine Learning Research* **26,** 1–39.

Douc, R. and Moulines, E. (2012). Asymptotic properties of the maximum likelihood estimation in misspecified hidden Markov models. *The Annals of Statistics* **40,** 2697–2732.

Douc, R., Moulines, E., Olsson, J., and van Handel, R. (2011). Consistency of the maximum likelihood estimator for general hidden Markov models. *The Annals of Statistics* **39,** 474–513.

Escola, S., Fontanini, A., Katz, D., and Paninski, L. (2011). Hidden Markov models for the stimulus-response relationships of multistate neural systems. *Neural Computation* **23,** 1071–1132.

Fox, E. B., Sudderth, E. B., Jordan, M. I., and Willsky, A. S. (2011). A sticky HDP-HMM with application to speaker diarization. *The Annals of Applied Statistics* **5,** 1020–1056.

Gimenez, O. (2026). *Bayesian Analysis of Capture-Recapture Data with Hidden Markov Models: Theory and Case Studies in R and NIMBLE*. CRC Press.

Glennie, R., Adam, T., and Leos-Barajas, V. (2023). Pitfalls and opportunities for hidden markov models in ecology. *Methods in Ecology and Evolution* **14,** 329–347.

Hooten, M. B. and Hobbs, N. T. (2015). A guide to Bayesian model selection for ecologists. *Ecological Monographs* **85,** 3–28.

Hooten, M. B., Johnson, D. S., McClintock, B. T., and Morales, J. M. (2017). *Animal*

Movement: Statistical Models for Telemetry Data. CRC Press.

Kozachenko, L. F. and Leonenko, N. N. (1987). Sample estimate of the entropy of a random vector. *Problemy Peredachi Informatsii* **23,** 9–16.

Langrock, R., King, R., Matthiopoulos, J., Thomas, L., Fortin, D., and Morales, J. M. (2012). Flexible and practical modeling of animal telemetry data: hidden markov models and extensions. *Ecology* **93,** 2336–2342.

Langrock, R. and Zucchini, W. (2011). Hidden Markov models with arbitrary state dwell-time distributions. *Computational Statistics & Data Analysis* **55,** 715–724.

Leos-Barajas, V., Photopoulou, T., Langrock, R., Patterson, T. A., Watanabe, Y. Y., Murgatroyd, M., et al. (2017). Analysis of animal accelerometer data using hidden Markov models. *Methods in Ecology and Evolution* **8,** 161–173.

Leroux, B. G. (1992). Maximum-likelihood estimation for hidden Markov models. *Stochastic Processes and their Applications* **40,** 127–143.

Li, J., Lai, M., Nguyen, N., Paschalidis, I. C., and Huggins, J. H. (2026). Robust model selection for discovery of latent mechanistic processes. Submitted manuscript.

Mardia, K. V. and Jupp, P. E. (2000). *Directional Statistics*. John Wiley & Sons, Chichester.

McClintock, B. T. and Michelot, T. (2020). Uncovering ecological state dynamics with hidden markov models. *Ecology Letters* **23,** 1878–1903.

Miller, J. W. and Dunson, D. B. (2019). Robust Bayesian inference via coarsening. *Journal of the American Statistical Association* **114,** 1113–1125.

Morales, J. M., Haydon, D. T., Frair, J., Holsinger, K. E., and Fryxell, J. M. (2004). Extracting more out of relocation data: Building movement models as mixtures of random walks. *Ecology* **85,** 2436–2445.

Patterson, T. A., Parton, A., Langrock, R., Blackwell, P. G., Thomas, L., and King, R. (2017). Statistical modelling of individual animal movement: An overview of key methods and a discussion of practical challenges. *AStA Advances in Statistical Analysis* **101,** 399–438.

Pohle, J., Langrock, R., van Beest, F. M., and Schmidt, N. M. (2017). Selecting the number of states in hidden markov models: pragmatic solutions illustrated using animal movement. *Journal of Agricultural, Biological and Environmental Statistics* **22,** 270–293.

Rabiner, L. R. (1989). A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE* **77,** 257–286.

Schwarz, G. (1978). Estimating the dimension of a model. *The Annals of Statistics* **6,** 461–464.

Vehkaoja, A., Somppi, S., Törnqvist, H., Valldeoriola Cardó, A., Kumpulainen, P., Väät äjä, H., et al. (2025). Movement sensor dataset for dog behavior classification. `https://doi.org/10.17632/vxhx934tbn.4`.

Wang, Q., Kulkarni, S. R., and Verdú, S. (2009). Divergence estimation for multidimensional densities via k-nearest-neighbor distances. *IEEE Transactions on Information Theory* **55,** 2392–2405.

Xue, C., Miller, J. W., Carter, S. L., and Huggins, J. H. (2026). Robust discovery of mutational signatures using power posteriors. *PLOS Computational Biology* **22,** e1014372.

Zucchini, W., MacDonald, I. L., and Langrock, R. (2016). *Hidden Markov Models for Time Series: An Introduction Using R.* Chapman and Hall/CRC, 2 edition.

Supplementary Material

Web Appendices A–C and Web Figures 1–8 referenced in Sections 3 and 4 are available with this paper at the *Biometrics* website.

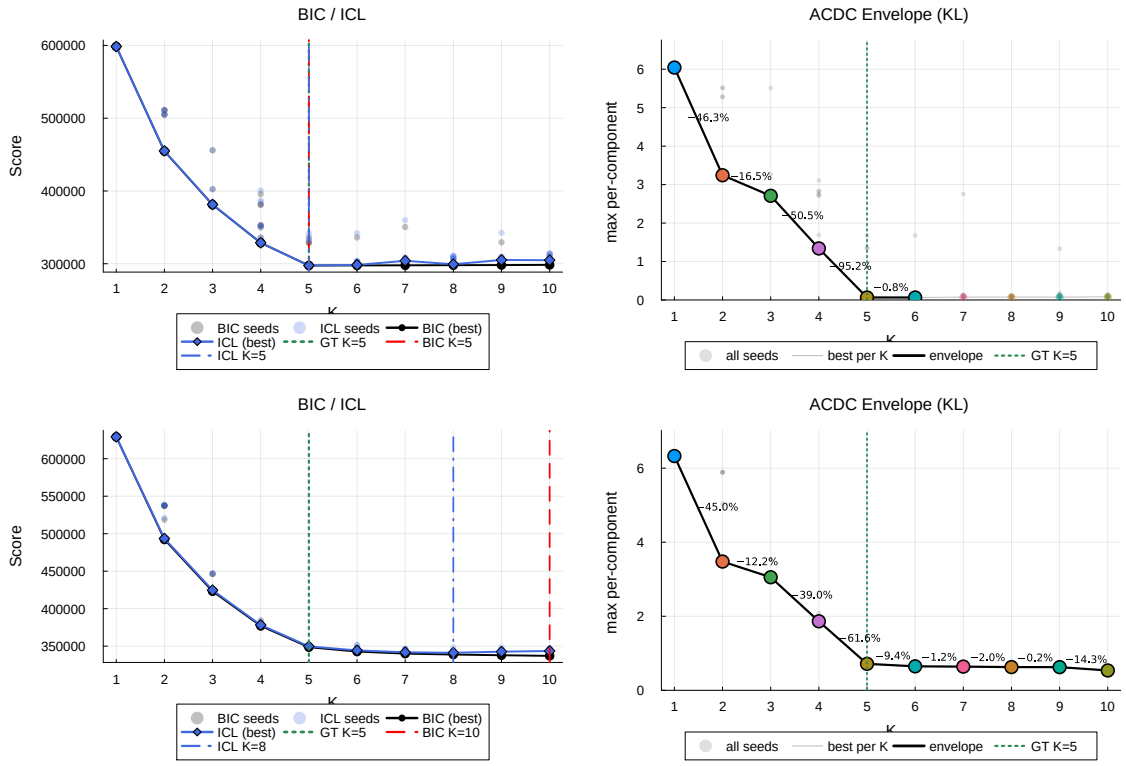
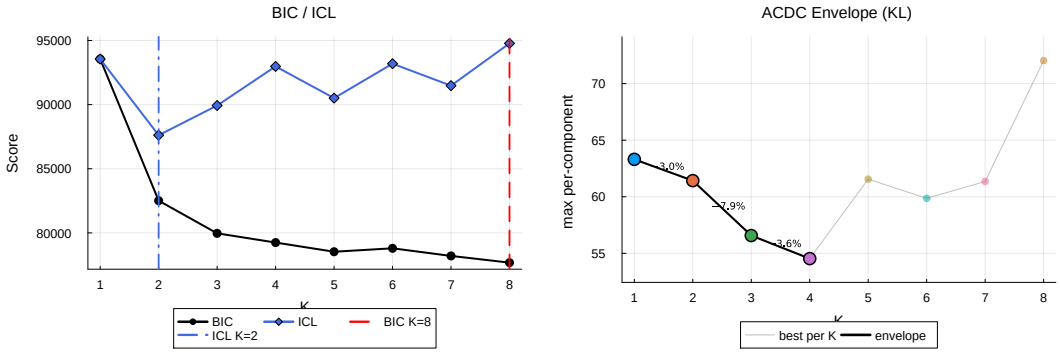
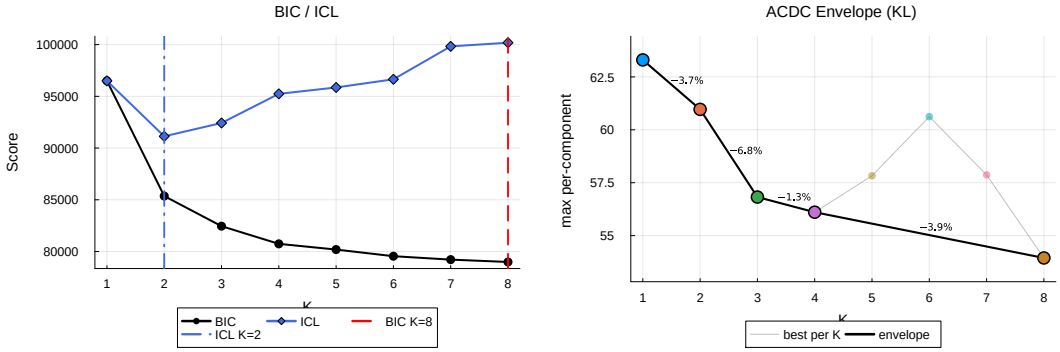


Fig 1: KL-based model-selection diagnostics for the synthetic spike-train HMM under well-specified Poisson emissions (top) and misspecified overdispersed contaminated data fit with a Poisson HMM (bottom). In the well-specified case, BIC, ICL, and ACDC recover the ground-truth five-state model. In the misspecified case, BIC and ICL overfit, although ICL is less aggressive, while ACDC retains the correct five-state recommendation.

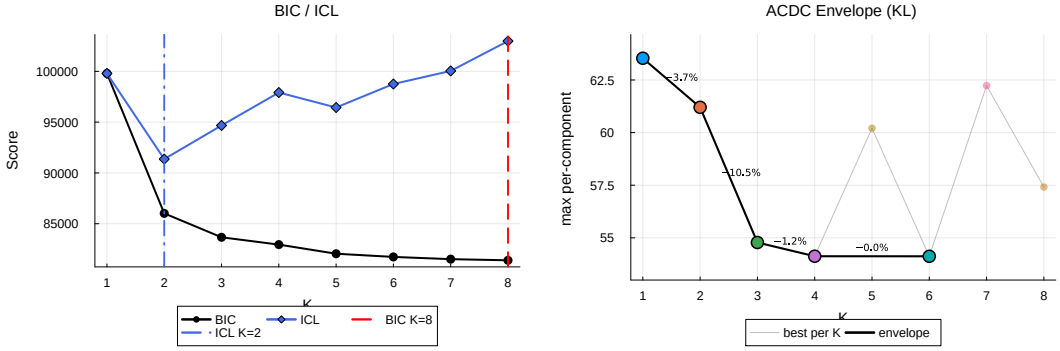
Alt text: Two rows compare fitted state numbers from one to ten. Each row shows BIC and ICL scores on the left and the ACDC maximum per-component discrepancy on the right. In the well-specified data, all three criteria identify five states. With overdispersion and burst contamination, BIC selects ten states and ICL selects eight, while the ACDC discrepancy drops sharply through five states and then nearly levels off.



(a) Wrapped Cauchy angle



(b) Gaussian raw angle



(c) Cardioid angle

Fig 2: Model-selection diagnostics for the elephant movement HMM under three angular emission specifications. In every case, BIC favors the largest tested K, ICL favors a coarse two-state split, and ACDC identifies an early, stable three-state decomposition; the Gaussian raw-angle model additionally produces a secondary high-K ACDC improvement consistent with splitting circular structure near the $-\pi/\pi$ boundary.

Alt text: Three vertically stacked sets of model-selection plots represent wrapped Cauchy, Gaussian raw-angle, and cardioid angle emissions. Each set compares BIC and ICL scores with the ACDC discrepancy across one to eight states. BIC continues to favor eight states, ICL favors two, and ACDC shows its main improvement by three states; the Gaussian raw-angle plot also shows a smaller improvement at eight states.

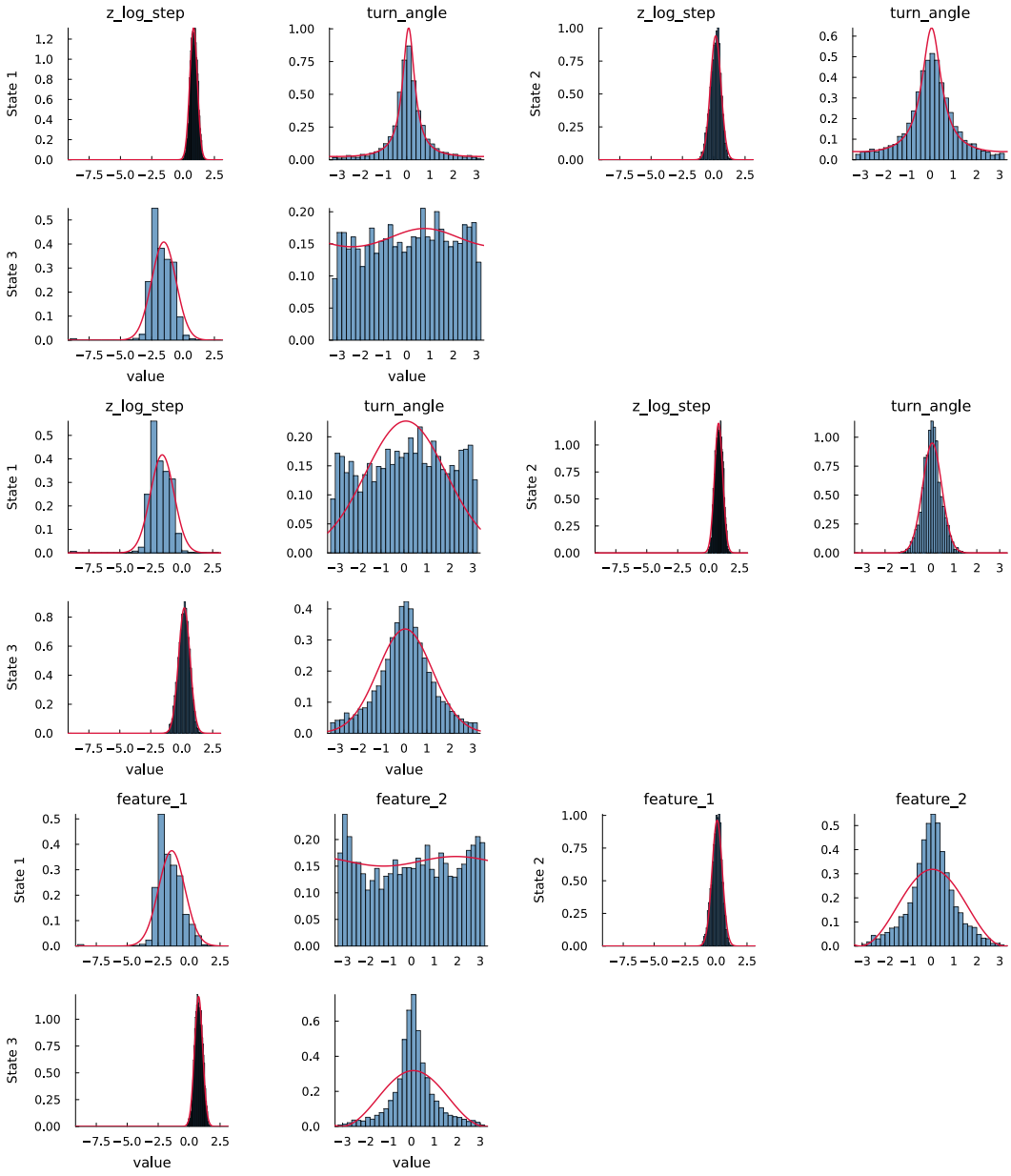


Fig 3: State-feature summaries for the $K = 3$ elephant movement decomposition under the three emission specifications: wrapped Cauchy angle model (top), Gaussian raw-angle model (middle), and cardioid angle model (bottom). Despite differences in angular modeling assumptions, the fitted states correspond to the same qualitative movement regimes: near-stationary movement, medium-speed movement, and fast directed travel.

Alt text: Three rows of state-specific histograms, one row for each angular emission model, compare log step length and turn angle for the three inferred states. Blue bars show empirical distributions and red curves show fitted emissions. Across models, the log-step distributions separate near-stationary, medium-speed, and fast movement, while the turn-angle shapes vary with the emission specification.

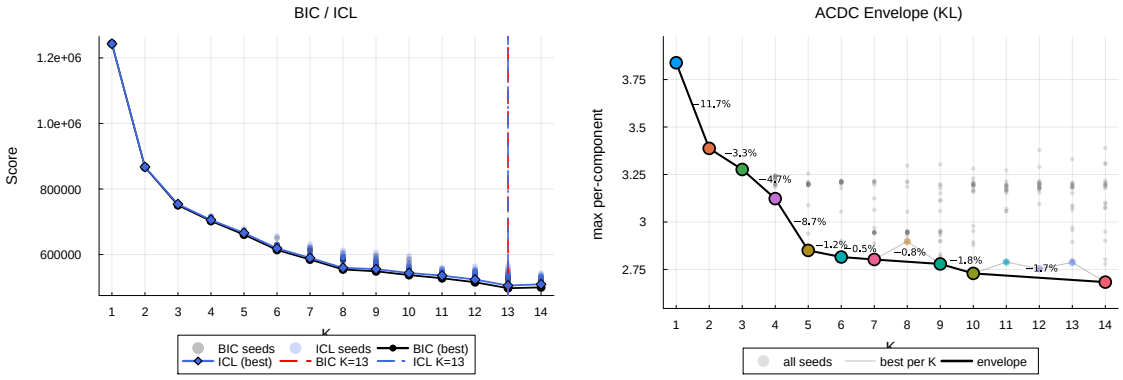


Fig 4: Model-selection diagnostics for the multi-dog bout-HMM across candidate state numbers. BIC and ICL select $K = 13$, while the KL-based ACDC envelope has its clearest elbow at $K = 5$; later envelope improvements are small and are interpreted as refinements or residual fitting.

Alt text: Two plots compare candidate models with one to fourteen states. BIC and ICL scores on the left decline toward their minimum at thirteen states. The ACDC maximum per-component discrepancy on the right falls most noticeably through five states and changes only slightly for larger models.

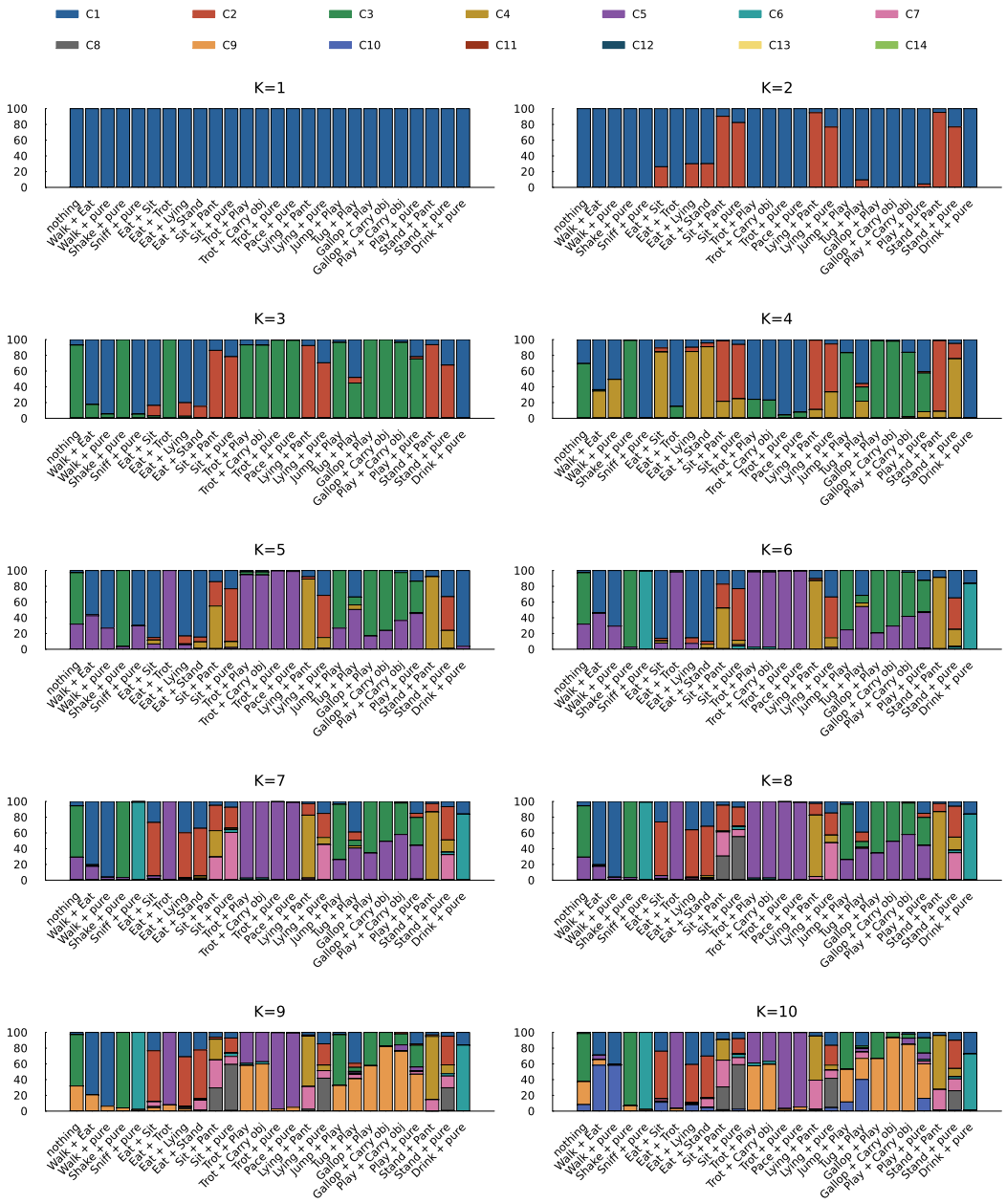


Fig 5: Behavior-pair distributions for aligned latent states in the multi-dog bout-HMM across candidate state numbers $K = 1, \ldots, 10$, using the ACDC-KL-best fit at each K. The $K = 5$ elbow fit separates the main sensor regimes, while larger models introduce finer redistributions such as cleaner subtyping of head-down, walking, and still-posture windows.

Alt text: Ten panels, one for each model from one to ten states, show stacked bars for the distribution of latent states across paired dog-behavior annotations. A single state accounts for every annotation pair in the one-state panel. By five states, several distinct color patterns separate broad behavior groups; larger models split those patterns into finer subgroups.

TABLE 1

Ground-truth state parameters for the synthetic spike-train HMM. Each state specifies a five-dimensional vector of neuron-specific Poisson firing rates; the reported occupancy is the realized fraction of bins assigned to each state in the simulated sequence.

State	Label	Firing rates across five neurons (Hz)	Empirical occupancy
1	Silent	$(2, 2, 2, 2, 2)$	20.8%
2	Chaotic	$(2, 25, 80, 2, 25)$	18.6%
3	Up	$(25, 30, 20, 35, 25)$	23.2%
4	Burst A	$(80, 5, 80, 5, 80)$	19.7%
5	Burst B	$(5, 80, 5, 80, 5)$	17.7%

TABLE 2
Plain-language summary of the $K = 5$ dog bout-HMM identified by the ACDC envelope.

State	Main annotations	Interpretation
C1	Drinking, sniffing, walking, eating with posture	Head-down or slow movement windows.
C2	Sitting, lying, standing	Mostly still posture windows.
C3	Shaking, galloping, jumping, play	Short bursts of strong whole-body motion.
C4	Standing, lying, sitting with panting	Still posture windows with panting annotations.
C5	Trotting, pacing, galloping, active play	Fast gait or active play windows.